MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-338( )/TRC-121

This specification is approved for use by all Departments and Agencies of the DOD.

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of Radio Terminal Set AN/TRC-121( ) which is capable of transmitting and receiving two 12 or 24 channel pulse coded multiplexed base band transmission over troposcatter radio paths. (See 6.1 and 6.5).

1.2 Description.- Shelter, Electrical Equipment S-338( )/TRC-121 hereinafter called the shelter facility, includes a modified S-280( )/G Shelter and all racks, brackets, shelves, tie-downs, wave-guides, other electrical components and wiring to enable installation and operation of the assemblage components of the AN/TRC-121( ). (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC 5820

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys & Titanium, Spot & Seam
MIL-T-7928	Terminals, Lug & Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-C-55427	Cable Assembly, Radio Frequency General Specification For
MIL-P-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging & Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual & Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

DL-SC-B-602302	Shelter, Electrical Equipment S-338( )/TRC-121
SC-D-602477	Power Schematic S-338( )/TRC-121
SC-D-602485	Schematic Control Indicator Unit
SC-F-602502	Power Wiring Diagram

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SC-D-602520	Waveguide & RF Schematic
SC-D-602521	Control Wiring Diag System I-R.S. AN/TRC-121
SC-D-602522	Control Wiring Diag System II-C.S. AN/TRC-121

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings listed on DL-SC-B-602302. (See 4.6)

3.3 Electrical Installation.- (See 3.22)

3.3.1 Power System Wiring and Components.- (See 4.7.1) Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system and wiring components) constituting a four-wire (115/230 V AC, plus ground electrical power distribution system) shall be in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of Conductors.- Electrical continuity shall exist between each pin of the power entrance connectors and respective circuit terminations thereof when tested in accordance with 4.7.1 and 4.7.1.1. The resistance shall be less than 3 ohms.

3.3.1.2 Dielectric Breakdown Voltage.- The insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

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3.3.1.3 Insulation Resistance.- The resistance of insulation between any two circuit conductors and also between each of the three ungrounded power-line conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

3.3.2 Power Distribution and Operational Functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown by Drawing SC-D-602477 when tested in accordance with 4.7.2.

3.3.3 Signal (Audio) and Control Wiring and Components.- (See 4.7.3) Installation and wiring of signal and control wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be in accordance with the drawings. The completed in-place signal (audio) and control system installation shall meet the following requirements:

3.3.3.1 Continuity.- There shall be circuit continuity between end terminations of all audio signal and control conductors when tested in accordance with 4.7.3 and 4.7.3.1. The resistance shall be less than 5 ohms.

3.3.3.2 Dielectric and Insulation Resistance Tests.- The insulation of all signal wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds and the insulation resistance shall not be less than 100 megohms when tested in accordance with 4.7.3.2.

3.3.4 Video and Radio Frequency (RF), System Cabling and Components.- Installation of video and RF cable assemblies, and connecting components (including plugs, connectors, adapters, fittings, receptacles, and lightning protectors) shall be in accordance with the drawings. The completed in-place video and RF system installation shall meet the following requirements:

3.3.4.1 Continuity of Conductors.- The point-to-point resistance between the following video and RF cable terminations shall be less than 1 ohm when tested in accordance with 4.7.4 and 4.7.4.1.

a. Both end terminations of the inner conductor.

b. Both end terminations of the outer conductor (shield).

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3.3.4.2 Dielectric and Insulation Resistance Tests.- The insulation of all video and RF cables, components and lightning protectors shall withstand the following potentials with the insulation resistance not less than 100 megohms when tested in accordance with 4.7.4 and 4.7.4.2.

a. 1500 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements removed.

b. 1100 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements in place.

c. 1500 volts DC applied continuously for not less than 30 seconds to all RF transmission lines.

3.3.5 Waveguide Transmission Assemblies and Components.- The installation of waveguides, waveguide switch, directional couplers, dummy load and fittings constituting the 4.4-5.0 GHz transmission system shall be in accordance with the drawings. The completed in-place installation shall meet the following requirements:

3.3.5.1 Voltage Standing Wave Radio (VSWR), Waveguide Assemblies .- The VSWR of each complete in-place waveguide transmission assembly shall not exceed 1.20 at the frequencies of 4.4,4.7 and 5.0 GHz when tested in accordance with 4.7.5 and 4.7.5.1.

3.3.5.2 Power Input, Transmit Waveguide Assembly.- With the shelter waveguide port (outside) terminated in a suitable dummy load, the transmit waveguide assembly shall withstand an input power level of not less than 1,000 watts at the frequencies of 4.4 and 5.0 GHz without arcing. (See 4.7.5 and 4.7.5.2)

3.3.5.3 Insertion Loss. The insertion loss (forward power transmission loss) between each adjacent port, shall not exceed 1.25 db at frequencies of 4.6 and 4.8 GHz when tested in accordance with 4.7.5.3.

3.3.6 Control and Indicator Units Operational Functions.- Switching, control and indicating functions of the control and indicator unit assemblies shall meet the operational requirements as denoted by Drawing SC-D-602485. (See 4.7.6)

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3.3.7 Cable Assemblies.- (See 4.4)

3.3.7.1 Cable Assemblies, Power and Audio Frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull tests and requirements of MIL-C-3885.

3.3.7.2 Cable Assemblies, Radio Frequency.- Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings and meet the following requirements when tested in accordance with MIL-C-55427:

3.3.7.2.1 DC Resistance.- When tested as specified, the point to point resistance of the conductor shall not be greater than 1 ohm.

3.3.7.2.2 Pull-out Load for Terminations.- When tested as specified the cable assembly shall show no evidence of mechanical damage when a load of 25 pounds is applied.

3.3.7.2.3 Dielectric Withstanding Voltage.- When tested as specified the cable assembly shall show no evidence of breakdown with a test potential of 1500 volts RMS, 25-100 hertz.

3.3.7.2.4 Insulation Resistance.- When tested as specified the insulation resistance shall not be less than 100 megohms with a test potential of 500 volts DC.

3.3.7.2.5 Voltage Standing Wave Radio.- When tested as specified the VSWR shall not exceed 1.20.

3.3.8 Suppression of Electromagnetic Interference.- The motors in the ventilating equipment used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4)

3.4 System Operation.- With all operational equipments of the AN/TRC-121( ) assemblage installed, the operational requirements of all power, audio, control and radio frequency cable assemblies, switches, connectors, waveguide assemblies, and control and indicator unit, shall be met when tested in accordance with 4.8.

3.5 Vehicular Transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.9)

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3.6 Rail transportaion.- The shelter facility, without any extra tieing, taping, etc., shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be subjected to and withstand the test of paragraph 4.10 without sustaining any buckling, cracking, delamination, or other permanent deformation.

3.7 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.11.1 and 4.11.2 there shall be no buckling, cracking, delamination or other permanent deformation.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the shelter facility or its walls, floor, roof, or door.

3.9 Fording.- The shelter facility shall be capable of withstanding the tests of 4.13. During and after the test there shall be no leakage into the shelter facility or its walls, floor, or door.

3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage or permanent distortion to the shelter facility mounting surface or insert head or threads and the insert shall not have rotated from its original position. Insert threads shall be free of paint.

3.11 Cleaning.- (See 3.22)

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

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3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.13 Marking.- (See 3.22 and 4.4)

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.13.2 Nameplates.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the first article.

3.13.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.14 Riveting.- (See 3.22) Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal, vapor- (See 3.22) All joints and raw edges of the framing for the power, signal, video, and antenna entrance boxes, air-conditioner, and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE-1204 as made by General Electric Company or Silastic 732 as made by Dow Corning, or equal, using the manufacturer recommended primer.

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3.16 Securing of parts.- (See 3.22) Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.- (See 3.22)

3.17.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.- (See 3.22) Canvas, webbing, wood, etc., shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

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3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (see 6.2e)

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.22 and 4.4)

3.20 Wiring and cabling.- (See 3.22) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

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3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Solderless Terminal Lug Connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of MIL-T-7928. (See 4.15)

3.20.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.16) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.17)

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3.2 Construction
3.3 Electrical installation
3.10 Mounting inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal, vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.18)

4.3 First Article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

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4.3.1 First article units.- The contractor shall furnish one (1) first article unit of the complete S-338( )/TRC-121.

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.21	4.16
3. Mounting Insert	3.10	4.14.1
4. Solderless-terminal-lug connectors	3.20.4	4.15
5. Visual and Mechanical	3.22	4.17
6. Power systems and components installation	3.3.1	4.7.1
7. Power distributional and operational function	3.3.2	4.7.2
8. Signal (audio) system wiring and components installation	3.3.3	4.7.3
9. Video and radio frequency (RF), system wiring and components installation	3.3.4	4.7.4
10. Waveguide transmission assemblies and components	3.3.5	4.7.5
11. Control and indicator units operational functions	3.3.6	4.7.6
12. System operation	3.4	4.8
13. Vehicular transportation	3.5	4.9
14. Rail transportation	3.6	4.10
15. Flat drop	3.7	4.11.1
16. Rotational drop	3.7	4.11.2
17. Watertightness	3.8	4.12

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Table I First Article Inspection (continued)

Inspections	Rqr Para	Test Para
18. Fording	3.9	4.13
19. Repeat 6. Power systems and components installation	3.3.1	4.7.1
20. Repeat 7. Power distributional and operational function	3.3.2	4.7.2
21. Repeat 8. Signal (audio) system wiring & components installation	3.3.3	4.7.3
22. Repeat 9. Video and radio frequency (RF) system wiring and components installation	3.3.4	4.7.4
23. Repeat 10. Waveguide transmission assemblies and components	3.3.5	4.7.5
24. Repeat 11. Control and indicator units operational functions	3.3.6	4.7.6
25. Repeat 12. System operation	3.4	4.8

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Cable assemblies	3.3.7
Suppression, RFI	3.3.8
Finish	3.12
Marking	3.13
Welding	3.19

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Visual and Mechanical	3.22	4.17	1.0	4.0
Electrical installation				**
Power systems and component installation				
Continuity of conductors	3.3.1.1	4.7.1.1		
Dielectric breakdown voltage	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Power distributional and operational function	3.3.2	4.7.2 thru 4.7.2.2		
Signal (audio) system wiring and components installation				
Continuity of conductors	3.3.3.1	4.7.3.1		
Dielectric and insulation resistance	3.3.3.2	4.7.3.2		
Video and radio frequency (RF), system wiring and components installation				
Continuity of conductors	3.3.4.1	4.7.4.1		
Dielectric & insulation	3.3.4.2	4.7.4.2		

** 1.0 for the entire group of power, signal and video wiring and waveguide assemblies

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Table II - Group A Inspection (Continued)

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Waveguide transmission assemblies and components				
Voltage standing wave ratio	3.3.5.1	4.7.5.1		
Power input, transmit waveguide assembly	3.3.5.2	4.7.5.2		
Insertion loss	3.3.5.3	4.7.5.3		
Control and indicator units operating function	3.3.6	4.7.6		

** All defects are major

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL % Def (See 4.5.2.1)
Interchangeability	3.21	4.16	4.0
Mounting insert	3.10	4.14.2	4.0
Watertightness	3.8	4.12	2.5
Fording	3.9	4.13	2.5

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NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV. Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.4.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Construction verification	3.2	4.6
Solderless-terminal-lug connection	3.20.4	4.15
Mounting inserts	3.10	4.14.1
System operation	3.4	4.8
Vehicular Transportation	3.5	4.9
Rail Transportation	3.6	4.10
Drops, Flat and Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3 The order of inspection within Group C for all inspections listed shall be as listed in Table IV. Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.12 and 4.13.

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4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6, shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall be performed as the selected unit is being assembled.

4.5.3.2 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of drop tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the vehicular and rail transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts
b. System operation
c. Truck transportability
d. Railroad transportability

4.5.3.5 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.4)

4.5.5 Alternate Group A audit and Group B inspection procedures.- The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with continuous sampling multilevel plans described in MIL-STD-1235. Sampling plans for the AQL's listed in Table II Group A inspection, and Table III Group B inspection of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

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b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government Verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

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4.7 Electrical installation tests.- The tests of 4.7.1 through 4.7.6 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115/230 volts AC, 60 cycles, single phase, and 24 volts DC.

4.7.1 Power systems and components installation tests.- (see 3.3.1) Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2, and 4.7.1.3, the following shall be accomplished. (Refer to Drawings SC-D-602477 and SC-F-602502).

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon (power indicator and power distribution box) lamps.

(2) All fluorescent lamp starters.

(3) Blowers and heaters.

(4) Control indicator units power plugs

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so as not to short to each other or to shelter facility ground.

4.7.1.1 Continuity of conductors.- Continuity resistance tests to meet the requirements of 3.3.1.1 shall be performed on each power circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position.

(1) LIGHT-1 switch

(2) LIGHT-2 switch

(3) Incandescent Cold Start switch

(4) BLOWER-1 switch

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(5) BLOWER-2 switch

(6) Main circuit breaker

(7) All branch circuit breakers

b. Place BLACKOUT/BYPASS switch to the BYPASS position.

c. Measure continuity as follows:

FROM	TO
(1) Pin 1 of J2	Grounding contacts of all 115 VAC power receptacles
(2) Pin 1 of J2	Ground rod terminal, E1
(3) Pin 1 of J2	Pin F of power amplifier plugs
(4) Pin 1 of J2	Pin D of control indicator's plug
(5) Pin 1 of J2	Grounding terminal in Power Distribution Box
(6) Pin 2 of J2	Brass contacts of 115 VAC power receptacles J5, J8, J9 and J13
(7) Pin 2 of J2	Pin c of power amplifier plugs
(8) Pin 2 of J2	Pin B of control indicator's plug
(9) Pin 2 of J2	Black wire terminations at each incandescent and fluorescent lamp
(10) Pin 3 of J2	Silver contacts of all 115 VAC power receptacles
(11) Pin 3 of J2	Pin B of power amplifier receptacles

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FROM	TO
(12) Pin 3 of J2	Pin C of control indicator's plug
(13) Pin 3 of J2	Silver contacts of all 115 VAC power receptacles
(14) Pin 3 of J2	White wire terminations at each incandescent and fluorescent lamp
(15) Pin 4 of J2	Brass contacts of 115 VAC power receptacles J4, J6, J7, J10, J11, J12, J15, J16, and J17
(16) Pin 4 of J2	Pin A of power amplifier plugs
(17) Pin 4 of J2	Pin A of control indicator's plug

d. Place the 24 volt DC power transfer switch in the power supply position. Measure continuity from the "TO AN/GRC-106" black jack to "TO DC Converter" black (-) conductor. Repeat continuity test from the red jack to red (+) conductor.

e. Place the 24 volt DC power transfer switch in the BATTERY position. Measure continuity from Pin B of 24 volt DC receptacle to "TO AN/GRC-106" black jack. Repeat continuity test from Pin A to the red jack.

f. Continuity tests measurements of steps c, d, and e shall meet the requirements of 3.3.1.1.

4.7.1.2 Dielectric Strength of Insulation.- Dielectric strength of insulation tests shall be made on each 60 cycle power line circuit conductor, including the neutral and each 24 volt DC Conductor. (See 3.3.1.2). A test voltage of 500 volts DC shall be applied for 30 seconds to the conductor under test with all other conductors placed at ground potential. Dielectric tests shall be made by applying the test voltage between ground and terminals of 120/240 volts AC Power Entrance Receptacle, J2 and 24 volts DC Receptacle, J1 as follows:

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a. Pin No. 2 of J2 (Pins No. 3 and No. 4 Grounded).

b. Pin No. 3 of J2 (Pins No. 2 and No. 4 Grounded).

c. Pin No. 4 of J2 (Pins No. 2 and No. 3 Grounded).

d. Pin No. A of J1 (Pin B Grounded).

e. Pin No. B of J1 (Pin A Grounded)

4.7.1.3 Resistance of Insulation.- Insulation resistance tests shall be made on each power line conductor. (See 3.3.1.3). Test conditions, preparation, points of measurement and grounding of conductors shall be as given for 4.7.1.2 above. The insulation resistance of all conductors with respect to ground shall exceed 100 megohms.

4.7.2 Power Distribution and Operational Function Tests.- Power distribution and operational function tests shall be made subsequent to the tests of 4.7.1 and with all electrical sources and loads connected except as given by 4.7.2.(e). All components which are part of and/or associated with the electrical power system shall be in-place and connected. Prior to conducting the tests of 4.7.2.1 and 4.7.2.2, the following shall be accomplished (Refer to Drawing SC-D-602477):

a. Secure blower exhaust port covers in OPEN position.

b. Open shelter door.

c. Place all circuit breakers (including main) on the 60 cycle 115/230 volts power distribution panel in the OFF position.

d. Provide and place suitable test lamps or equivalent test indicators in each 6o cycle power outlet receptacle.

e. Disconnect Plug from Control Indicator Units.

f. Provide and connect a suitable 120 volt test lamp or test indicator between Pins A and C and 240 volt test lamp or indicator between Pins A and B of Control Indicator Units.

g. Provide and connect a suitable 120 volt test lamp or test indicator between Pins B and C and 240 volt test lamp or indicator between Pins A and C of power amplifier plugs.

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h. Place all the lighting control switches in the ON position.

i. Place the blackout by-pass switches in the BYPASS position.

j. Set Thermo Switches to 170°F.

k. Provide and connect a 24 volt DC meter to the "To AN/GRC-106" jacks (Positive Meter Lead to Red Jack; Negative Meter Lead to Black Jack).

4.7.2.1 60 Cycle AC Power Distribution Tests.-

a. Apply 115/230 single phase 60 cycle power to connector in power entrance box. POWER ON Indicator Lamps shall light.

b. Rotate Meter Switch, S2 to Positions L1, L2 and OFF. The Voltmeter shall indicate specific line voltages and no voltage in the respective switches position.

c. Place main circuit breaker (CB-1) in ON position. Turn each branch circuit breaker ON in sequence given below. The corresponding indicator lamp shall light and the indication shall be as listed below for each breaker.

	Circuit Breaker No.	Indication
(1)	CB-2	Both test lamps connected to the curbside pwr amp plug shall light
(2)	CB-3	Both test lamps connected to the roadside pwr amp plug shall light
(3)	CB-4	Both test lamps connected to the curbside control indicator plug shall light
(4)	CB-5	Both test lamps connected to the roadside control indicator plug shall light

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Circuit Breaker No.	Indication
(5) CB-6	Test lamp in curbside transmitter receptacle shall light
(6) CB-7	Test lamp in roadside transmitter receptacle shall light
(7) CB-8	Test lamps in curbside receiver and converter receptacles shall light
(8) CB-9	Test lamps in roadside receiver and converter receptacles shall light
(9) CB-10	Both blowers shall operate and exhaust air from the shelter
(10) CB-11	All fluorescent and incandescent lamps shall light
(11) CB-12	Test lamp in heater-1 receptacle shall light
(12) CB-13	Test lamp in heater-2 receptacle shall light
(13) CB-14	No other indication
(14) CB-15	Test lamps in convenience receptacles, J15 thru J17, shall light
(15) CB-16	No other indication

d. With all circuit breakers remaining in the ON position, operate power controls switches to OFF position in sequence given below. The indication shall be as listed for each switch.

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Switch		Indication
(1)	LIGHT-1	Half of roadside and curbside fluorescent lamps shall extinguish
(2)	LIGHT-2	Remaining fluorescent lamps shall extinguish
(3)	Incandescent cold start	All incandescent lamps shall extinguish
(4)	Blower-1	Blower-1 shall stop
(5)	Blower-2	Blower-2 shall stop

e. With Blower No. 1 Switch in the OFF position, place a jumper across the contacts of Thermo Switch, S4. Blower No. 1 shall operate. Remove jumper; blower shall stop.

f. Repeat step e for Blower No. 2 and Thermo Switch, S6.

g. With all circuit breakers and 60 cycle power control switches in the ON position, perform the following lighting blackout test:

(1) Place blackout by-pass switch in the OFF position. Close the shelter door. All fluorescent and incandescent lamps shall light.

(2) With the blackout by-pass switch remaining in the OFF position, open the shelter door. All fluorescent and incandescent lamps shall go out.

h. Turn OFF the main and all branch line circuit breakers. Provide and connect a resistive load of 12 ohms (5 KW) between Terminals 1 and 2 of roadside terminal block. Turn ON main and branch circuit breakers, CB-1 and CB-5. Operate Meter Switch, S1 as given below. The indication shall be as listed.

Meter Switch Position		Indication
(1)	L1	Ammeter shall read 20 amps ± 5 Percent.
(2)	L2	Ammeter shall read 20 amps ± 5 Percent

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4.7.2.2 24 Volt Power Distribution Tests.-

a. Apply 24 volts DC to EXT Battery Receptacle with polarity as indicated (Negative to Pin B; Positive to Pin A).

b. Place 24 volt DC Transfer Switch in the EXT BATTERY position. Lamp, DS-23 shall indicate correct polarity and test meter (See 4.7.2(k)) shall read 24 volts.

c. Place 24 volts DC Transfer Switch in the DC CONVERTER position. Lamp, DS-23 shall go out and test meter shall read zero volts.

d. With DC Transfer Switch remaining in the DC CONVERTER position, connect 24 volt DC Power to the DC Converter conductors (Positive to Red Conductor; Negative to Black Conductor). Lamp, DS-23 shall indicate correct polarity and test meter shall read 24 volts.

e. Disconnect test meter and repeat test step d. above with reversed 24 volt DC polarity (Positive to Black Conductor; Negative to Red Conductor). Lamp, DS-23 shall indicate incorrect polarity.

4.7.3 Signal (Audio) and Control Wiring and Components Installation Tests.- Prior to conducting signal and control wiring and components tests of 4.7.3.1 and 4.7.3.2, the main Power Circuit Breaker (CB-1) shall be placed in the OFF position and following removed or disconnected (Refer to Drawings SC-D-602521 and SC-D-602522):

a. Plugs to Control-Indicator Unit.

b. Plugs to Waveguide Switch.

c. Power Amplifier's (AM-6090) Air Pre-Heating Element.

d. 1100 OHM Resistor connected between Terminals 14 and 15 and condenser between terminals 14 and 22 of TB-1. (Roadside and Curbside)

e. Connect jumper wires between terminals 14 and 22 of both TB-1.

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4.7.3.1 Continuity of Conductors.- Continuity tests shall be performed on each audio and control line circuit conductor, (See 3.3.3.1). A low resistance ohmmeter or equivalent continuity tester shall be used. Continuity measurements shall be made as follows:

	FROM (Roadside)	TO (Roadside)
a.	Pin a of P6	Pin B of P20
b.	Pin c of P6	Pin A of P20
c.	Pin F of P6	Pin E of P20
d.	Pin M of P6	Pin G of P20
e.	Pin N of P6	Pin H of P20
f.	Pin P of P6	Pin D of P20
g.	Pin R of P6	Pin C of P20
h.	Pin S of P6	Blk Cond, CV-425(TB-1-22)
i.	Pin T of P6	Wht Cond, CV-425 (TB-1-15)
j.	Pin Z of P6	Shield, CV-425 PR (TB-1-16)
k.	Pin K of P6	Pin A of P14
l.	Pin L of P6	Pin B of P14
m.	Pin U of P6	Pin D of P3
n.	Pin V of P6	Pin F of P3
o.	Pin H of P6	Pin B of P11
p.	Pin J of P6	Pin A of P11
q.	Pin W of P6	Pin C of P11
r.	Pin X of P6	Pin K of P11

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	FROM (Roadside)	TO (Roadside)
s.	Pin Y of P6	Pin K of P11
t.	Pin A of J1	Pin A of P13
u.	Pin B of J1	Pin B of P13
v.	Pin C of J1	Pin C of P13
w.	Pin D of J1	Pin D of P13
x.	Pin E of J1	Pin E of P13
y.	Pin F of J1	Blk CV-425 Cond (BP-1) and Binding Post No. 1
z.	Pin G of J1	Wht CV-425 Cond (BP-2) and Binding Post No. 2
aa.	Pin E of P21	Red Cond to P.A. Heater Element
bb.	Pin F of P21	Blk Cond to P.A. Heater Element
cc.	Pin C of P21	Pin A of P16
dd.	Pin G of P2	Pin C of P16
ee.	Pin B of P16	Pin E of P16
ff.	Pin F of P20	TML 7 of TB-1
gg.	Pin P of P20	TML 4 of TB-1
hh.	Pin N of P20	Pin D of P14
ii.	Pin J of P20	Pin A of P3
jj.	Pin K of P20	Pin C of P3
kk.	Pin L of P20	Pin B of P3 and Pin E of P14

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Repeat continuity checks of above for corresponding curbside signal lines; substituting P7 for P6, P22 for P20, P9 for P14, P4 for P11, P5 for P13, P23 for P21, P10 for P16, and P8 for P3.

4.7.3.2 Dielectric and Insulation Resistance Tests.- Dielectric and insulation resistance tests to meet the requirements of 3.3.3.2 shall be performed on each signal and control line circuit in accordance with the following paragraphs. The test voltage shall be applied between the conductor(s) under test and shelter facility ground terminal while all other signal and control conductors are placed at shelter ground by the following procedures.

a. Using appropriate test jigs or fixtures, connect the following to shelter facility ground terminal.

(1) All 19 conductor pin terminations of transmitter's plug P6 and P7.

(2) All 7 conductor pin terminations of roadside and curbside antenna sighting connectors J1.

(3) Conductor pin terminations E, F and G of control indicator's plugs P21 and P23.

(4) Conductor pin terminations F, J, K, L, M, N, and P of control indicator's plugs P20 and P22.

b. Remove ground connection from pin a of roadside transmitter plug P6, and connect tester between this pin and shelter ground terminal.

c. Adjust tester for 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met. Disconnect tester and re-establish ground connection to the pin under test.

d. Repeat steps b and c above for the following pin terminations; each conductor pin terminations (except pins Y and Z) of transmitter plug P6 and P7; each conductor pin terminations (except pin C) of each sighting connector J1; conductor pin terminations E, F and G of control indicator plugs P21 and P23; and conductor pin terminations F, J, K, L, M, N, and P of control indicator plugs P20 and P22.

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e. Connect tester between pin B of roadside pwr amp plug P16 and shelter ground. Repeat test of step c. Repeat for curbside pwr amp plug P10. Disconnect tester.

f. Remove temporary wire jumpers between terminals 14 and 22 of both TB-1 and reconnect condenser. Reconnect 1100 ohm resistors.

4.7.4 Video and Radio Frequency (RF) Cabling and Components Tests.- (Refer to Drawing No. SC-D-602520).

4.7.4.1 Continuity of Conductors.- Continuity tests to meet the requirements of 3.3.4.1 shall be performed on each video and RF coaxial line. A low resistance reading ohmmeter or equivalent instrument shall be used. Continuity testing shall be inner conductor to inner conductor and outer conductor (shield) to outer conductor (shield).

FROM	TO
TRAFFIC cable, XMTR CS*	TRAFFIC cable, RCVR CS
CS XMTR RF OUT	CS PWR AMP RF IN
DIR COUP CHAN A CS	XMTR CHAN A CS
DIR COUP CHAN B CS	XMTR CHAN B CS
TRAFFIC cable, XMTR RS**	TRAFFIC cable, RCVR RS
RS XMTR RF OUT	RS PWR AMP RF IN
DIR COUP CHAN A RS	XMTR CHAN A RS
DIR COUP CHAN B RS	XMTR CHAN B RS
TO CABLE RS	VID ENT BOX SYS-1 (Female)
FM CABLE RS	VID ENT BOX SYS-1 (Male)
TO CABLE CS	VID ENT BOX SYS-2 (Female)
FM CABLE CS	VID ENT BOX SYS-2 (Male)

*CS - Curbside
**RS - Roadside

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4.7.4.2 Dielectric and Insulation Resistance Tests.- Dielectric and insulation resistance tests to meet the requirements of 3.3.4.2 shall be performed on each video and RF circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

a. Remove all lightning protector elements at video entrance box.

b. Apply test potential of 1500 volts DC for not less than 30 seconds between the following pin connections and ground terminal. The requirements of 3.3.4.2 shall be met.

(1) System-1 male

(2) System-1 female

(3) System-2 male

(4) System-2 female

c. Replace lightning protector elements at video entrance box.

d. Repeat step b above applying test potential of 1100 volts DC.

e. Apply test potential of 1500 volts DC for not less than 5 seconds between the inner conductor and outer conductor (shield) of the following cable assemblies; CS XMTR TRAFFIC, CS XMTR RF OUT, CS DIR COUP CHAN A, CS DIR COUP CHAN B, RS XMTR TRAFFIC, RS XMTR RF OUT, RS DIR COUP CHAN A, and RS DIR COUP CHAN B. The requirements of 3.3.4.2 shall be met.

4.7.5 Waveguide Transmission Assemblies and Components Tests- The tests of 4.7.5.1 thru 4.7.5.3 shall be made on each in-place waveguide system. (Refer to Drawing SC-D-602520).

4.7.5.1 Voltage Standing Wave Ratio (VSWR).- The basic test set-up shall be as shown on Figure 1. Tuners and/or pads shall be included where necessary to reduce the residual VSWR of test set-up to a level which will assure accurate measurements. VSWR measurements shall be made as follows:

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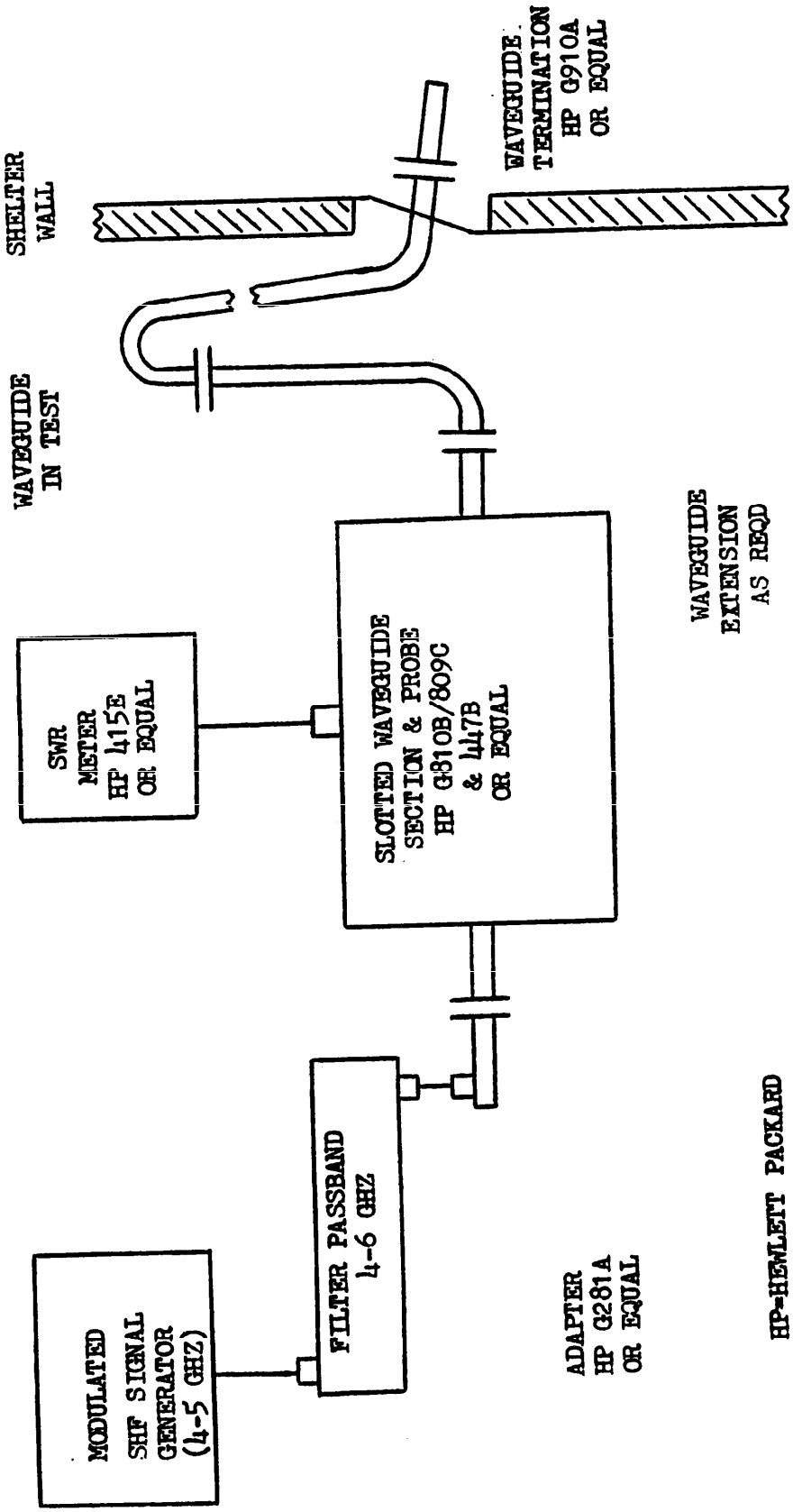


FIG. 1 VSWR TEST SET-UP FOR WAVEGUIDE ASSEMBLIES

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a. Connect waveguide termination (HP G-910A or equal) to the Roadside TRANSMIT antenna waveguide port in antenna entrance box.

b. Place the roadside facility waveguide switch in ANTENNA position.

c. Connect slotted waveguide section (thru extension waveguide as required) to the roadside AM-6090 power amplifier connecting waveguide flange.

d. Measure VSWR at frequencies of 4.4, 4.7, and 5.0 GHz. The requirements of 3.3.5.1 shall be met.

e. Place the facility waveguide switch in DUMMY load position.

f. Repeat step d above.

g. Connect waveguide termination to roadside A RECEIVE antenna waveguide port in antenna entrance box.

h. Connect slotted waveguide section to roadside R-1287 Receiver A channel connecting waveguide flange.

i. Repeat step d above.

j. Repeat steps g thru i above on roadside B receive waveguide assembly.

k. Repeat steps a thru j above for curbside waveguide assemblies.

4.7.5.2 RF Power Input Test, Transmit Waveguide Assembly.- RF power input test shall be conducted as follows:

a. Connect a high power (1,000 watts minimum) waveguide termination (dummy load) to the roadside TRANSMIT antenna waveguide port in antenna entrance box.

b. Connect a 1,000 watt RF (4.4-5GHz) power generator source, such as Power Amplifier, AM-6090 and suitable 150 MW exciter, to the roadside facility AM-6090 Power Amplifier connecting waveguide flange.

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c. Place the roadside facility waveguide switch in antenna position.

d. Adjust generator for 1,000 watt output at frequency of 4.4 GHz and apply power to the facility waveguide assembly for not less than 10 minutes. The requirements of 3.3.5.2 shall be met.

e. Repeat step d above at frequency of 5.0 GHz.

f. Place the facility waveguide switch in DUMMY LOAD position.

g. Repeat steps d and e above.

h. Repeat steps a thru g above for curbside waveguide assemblies.

4.7.5.3 Insertion Loss.- Insertion loss test to meet the requirement of 3.3.5.3 shall be performed on each waveguide assembly in accordance with the following paragraphs. A system such as shown in Figure 2 shall be used.

a. At 4.6 GHz, obtain a 0 db reference level by varying the source when connected as shown in "A" of Figure 2.

b. Tune for minimum reflection (VSWR of 1.0) by adjusting the slide screw tuner when connected as shown in "B".

c. Repeat process of steps a and b above to assure a 0 db reference level at unity VSWR.

d. Insert test item (roadside transmitter waveguide system) as shown in "C". The source is fed at the transmitter end; the load at the transmit antenna end.

e. Set waveguide at "TO ANT" position.

f. Read and record insertion loss directly from the calibrated power meter. The requirements of 3.3.5.3 shall be met.

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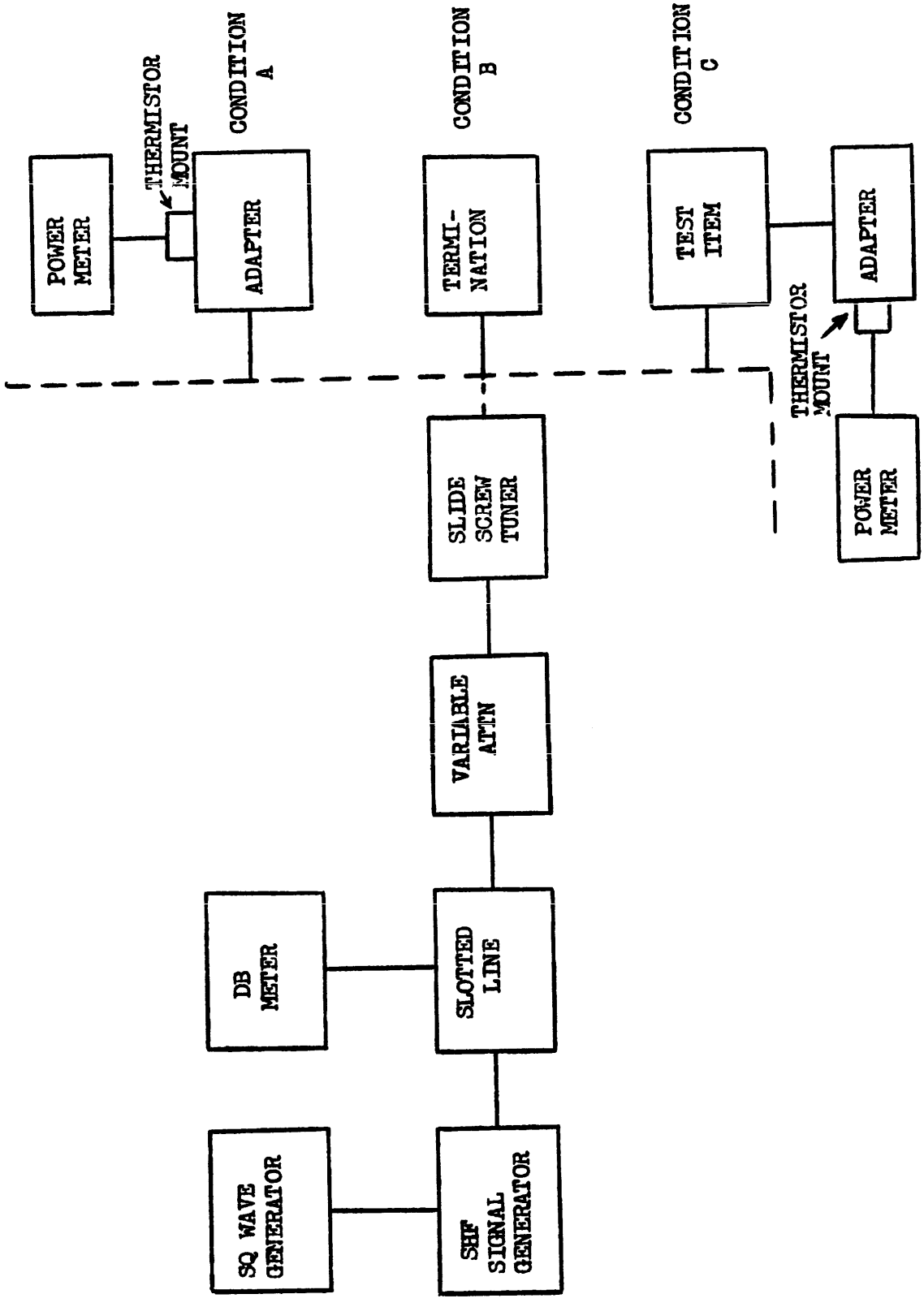


FIG. 2 INSERTION LOSS TEST SETUP

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g. With the test setup calibrated at 0 db reference and unity VSWR at 4.6 GHz, repeat steps d thru f above for the following waveguide systems:

Source Location	Load Location
Roadside ANT A	Roadside REC A
Roadside ANT B	Roadside REC B
Curbside XMTR	Curbside XMIT ANT
Curbside ANT A	Curbside REC A
Curbside ANT B	Curbside REC B

h. Repeat step a thru g above for a frequency of 4.8 GHz.

4.7.6 Control and Indicator Unit Operational Function Tests.- (See 3.3.6; reference Drawings SC-D-602485 and SC-D-602520).

a. Test Preparation

(1) Fabricate test fixture shown on Figure 3 and connect fixture to roadside facility control wiring terminations at connectors P6, P14 and P16 as indicated.

(2) Connect P20 and P21 to SYS-1 Control/indicator Unit.

(3) Connect a temporary jumper across the contacts of roadside thermal switch.

(4) Place AC power circuit breaker CB5 in OFF position.

(5) Place waveguide switch in the ANTENNA position.

(6) Set switch S1 of Control/indicator Unit in ALARM disable position.

(7) Set switches S1 and S2 of test fixture in OPEN position.

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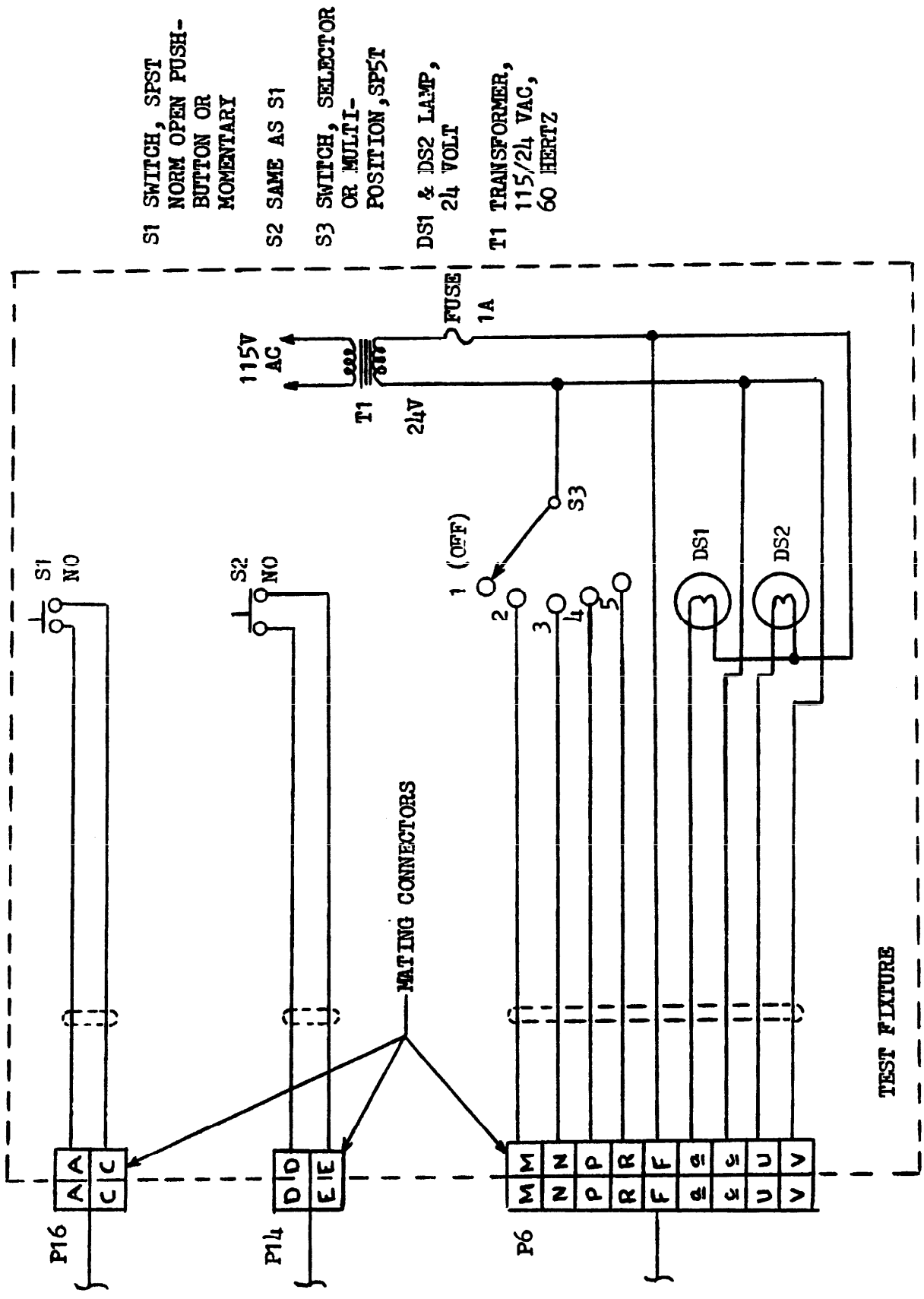


FIG. 3 SIGNAL/CONTROL TEST FIXTURE

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Power Distribution BOX

(8) Set switch S3 of test fixture to position OFF.

(9) Apply AC power to the test fixture.

b. Test Procedure -

(1) Apply 120/240 volts AC power to the Control/indicator Unit by placing circuit breaker CB5 in the ON position. The following SYS-1 indicator lamps shall glow:

Green lamps B and C of DS-1 on Control/indicator Unit (pre-htr off).

Green lamps A and B of DS-4 on Control/indicator Unit (ant).

Red lamps A*, B*, C and D of DS-5 on Control/indicator Unit (aud alarm disabled).

Lamp DS-2 on test fixture.

(2) Depress button of switch, S1 on Control/indicator Unit. Red lamps A, B, C and D of DS-5 on Control/indicator Unit (aud alarm disabled) shall extinguish and lamp, DS-1 on test fixture shall glow.

(3) Position waveguide switch mid-way between antenna and dummy load switch positions. Lamp DS-2 on test fixture and green lamps A and B of DS-4 of Control/indicator Unit (ant) shall extinguish.

(4) Place waveguide switch in the DUMMY load position. Red lamps D and C of DS-4 on Control/indicator Unit (Dum Load) and lamp DS-2 on test fixture shall glow.

(5) Rotate switch S3 on test fixture to Position 2. Red lamp A of DS-3 (traffic-in) on Control/indicator Unit shall glow.

(6) Rotate switch S3 on test fixture to Position 3. Green lamp D of DS-3 (traffic-in) on Control/indicator Unit shall glow.

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(7) Rotate switch S3 on test fixture to Position 4. Red lamp B of DS-3 (traffic-out) on Control/indicator Unit shall glow.

(8) Rotate switch S3 on test fixture to Position 5. Green lamp C of DS-3 (traffic-out) on Control/indicator Unit shall glow.

(9) Close switch S2 on test fixture. Red lamps A B, C* and D* of DS-2 on Control/indicator Unit (high-tem alert) shall glow.

(10) Close switch S1 on test fixture. Red lamps A and D* of DS-1 on Control/indicator Unit (pre-htr on) shall glow and power amplifier, AM-6090 air pre-heat elements shall get hot.

(11) With switch S1 on test fixture remaining closed, disconnect jumper across thermal switch, S11 (see 4.7.6a(3)). Red lamps A and D* of DS-1 on Control/indicator Unit (pre-htr on) shall extinguish; green lamps B and C of DS-1 (pre-htr off) shall glow; and, AM-6090 air pre-heat elements shall cool.

(12) Repeat steps (1) thru (11) above for curbside (SYS-2) facility control wiring.

*Intermittent (flashing) glow

4.8 Systems Operation Tests.- Install all operational equipment in their respective locations and connect power, signal and RF cable assemblies to their designated receptacles and connectors. (See 3.4) (See Figure 4 for block diagram).

a. Preliminary setup.- (The following applies to both systems of the AN/TRC-121)

(1) Position multiplexers TD-660/U and TD-204/U to facilitate connections to the video entrance box.

(2) Make the following cable connections between the TD-660/U and TD-204 using Cable Assemblies CG-1040B/U (5 ft)

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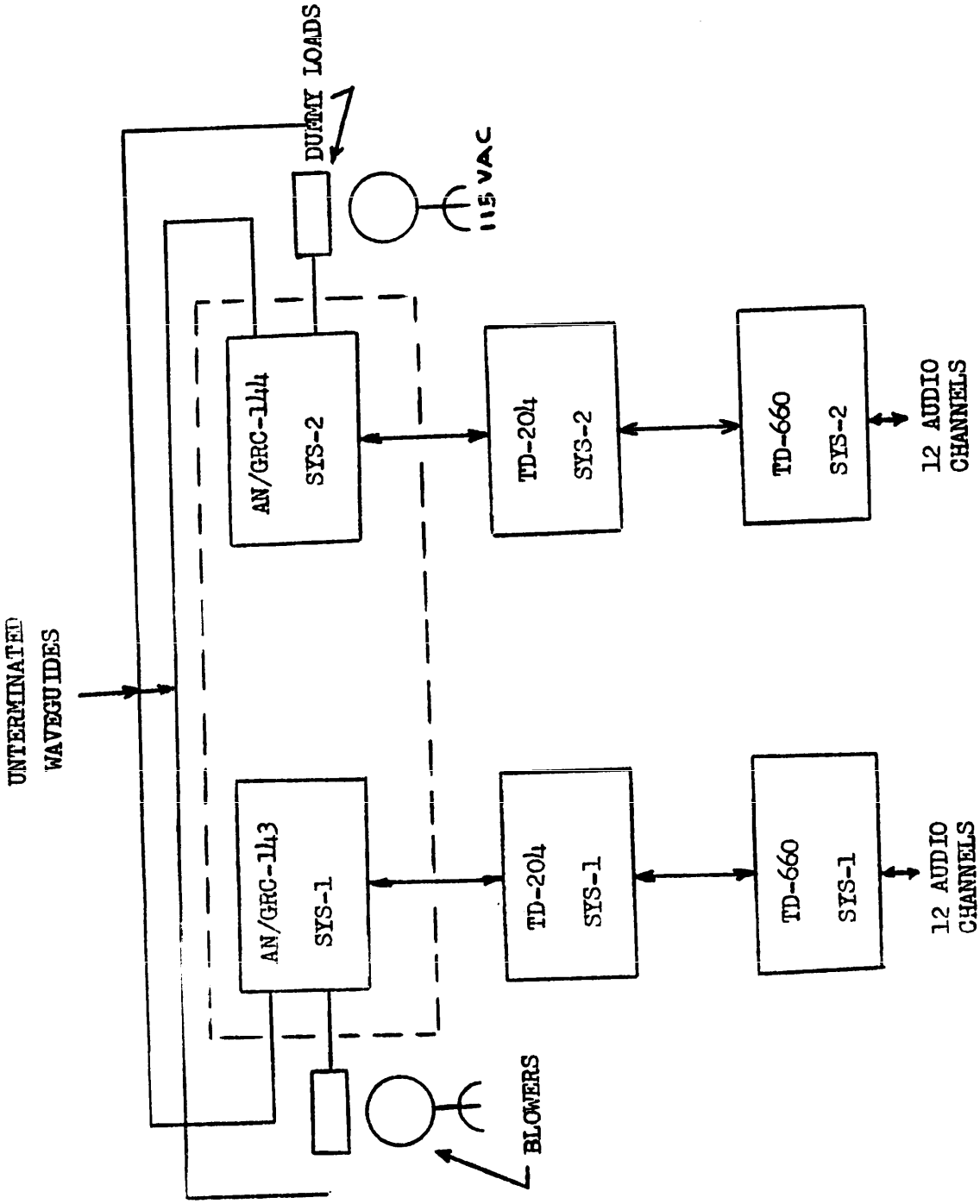


FIGURE 4 SYSTEM TEST SETUP

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TD-660/U	TO	TD-204/U
PCM IN		PCM OUT
PCM OUT		PCM IN
TIM IN		TIM OUT
TIM OUT		TIM IN

(3) Connect one end of cable assemblies CG-2437/TCC (15 ft) and CG-2438/TCC (15 ft) to the FROM CABLE and TO CABLE connectors of TD-204/U respectively.

(4) Connect other end of the CG-2437/TCC and CG-2438/TCC to cable assembly CX-10734( )/G.

(5) Connect remaining end of CX-10734( )/G to video connector at video entrance box.

(6) Connect Headset H-91A/U to the TD-204/U.

(7) Set 2 wire/4 wire switch of each TD-660/U to 2 wire, MASTER/SLAVE switch to MASTER, AUX switch to OUT, and switches I, II, III and IV to OFF.

(8) Set Traffic Sel switch of each TD-204/U to 12; READ/ZERO SET/NORM OPR switch to NORM OPR; MILE switches on panels 6A4 and 6A5 to 1/2; and Cable Current to OFF.

(9) Connect the shelter facility to an earth ground.

(10) Connect shelter facility to a 115/230 volt single phase AC power source.

(11) Open all shelter air intake and exhaust ports.

(12) Set power amplifiers air intake and exhaust ducts for external air intake and discharge.

(13) Place waveguide switches in dummy load position.

(14) Place all AC power circuit breakers ON.

(15) Turn on shelter air exhaust blowers.

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(16) Place the 24 volt AN/GRC-106 main power switch in the DC converter position.

(17) Connect a 400 watt RF dummy load, consisting of a 4 ohm non-inductive resistor in series with a 90 MMF condenser, between the AN/GRC-106 antenna terminal (whip) and grounding terminal of radio set.

(18) Set up Radio Set AN/GRC-143 for 12-channel operation (1/4 mile cable interface) by following appropriate procedures contained on the charts in each cabinet.

(19) Set up Radio Set AN/GRC-106 for CW operation on a frequency between 5.5 and 6.0 MHz.

(20) Turn dummy load blowers ON to reduce any heat build-up.

b. Operational test.-

(1) Tune radio sets AN/GRC-143 as indicated on the appropriate charts of each equipment. Use 4.6 GHz as SYS-1 receive and SYS-2 transmit frequencies; and 4.8 as SYS-1 transmit and SYS-2 receive frequencies.

(2) Set waveguide switches to TO ANT.

(3) Set multiplexers for proper operations.

(4) Depress the TALK-SIG switch on SYS-1 TD-204/U. The "ring" light on each transmitter shall turn amber and loud speaker sound 1600 Hz tone. The ring signal on SYS-2 TD-204/U shall be heard.

(5) Repeat step (4) above depressing TALK-SIG switch on SYS-2 TD-204/U.

(6) Press the RING pushbutton on SYS-1 transmitter. The RING indicator shall turn amber and a 1600 Hz signal shall be heard on both transmitters. The ring signal shall be heard on both external TD-204/U.

(7) Repeat step (6) pressing the RING pushbutton on SYS-2 transmitter.

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(8) Communicate over all phones connected to the radio sets and TD-204/U. Orderwire communication will also be heard over the loudspeakers.

(9) Connect Telephone Sets TA-312/PT to the remote orderwire connections at the video entrance box. Set CV-425's for proper operation.

(10) Ring TA-312/PT of system 1 from TA-312/PT of system 2; ring TA-312/PT of system 2 from TA-312/PT of system 1; establish communications between the phones.

(11) Connect Headsets H-91A/U to each TD-660/U. Establish communications between the two headsets on all 12 channels.

c. Test procedures for AN/GRC-106.-

(1) Set power amplifier meter selector switch to PWR OUT.

(2) With AN/GRC-106 Radio Set adjusted for full output power, depress button on handset and hold depress for 2 minutes. Observe PA meter indication. The power amplifier meter shall indicate full output power and remain steady during the 2 minute period.

4.9 Vehicular Transportation Test.- The facility shall be loaded to simulate the AN/TRC-121( ) using live and dummy loads (See 6.2b). The exhaust blowers and housings shall be mounted outside the shelter facility in their normal operating position. Using the lift and tie-down slings furnished with the bare shelter, the facility shall be secured as indicated on the shelter transportation data plate, to the bed of a government furnished 2-1/2 ton, M-35 truck. The tire pressure shall be set for cross-country travel as listed on the truck dashboard. The facility shall be centralized and blocked to prevent shifting. The facility shall be transported over the specially prepared course described in 4.9.1 for 10 laps at each speed of 5, 10, 15 and 18-20 MPH. A lap is defined as traversing the course in one direction. Upon completion of this test, the facility shall not have sustained any of the types of damage cited in 3.5.

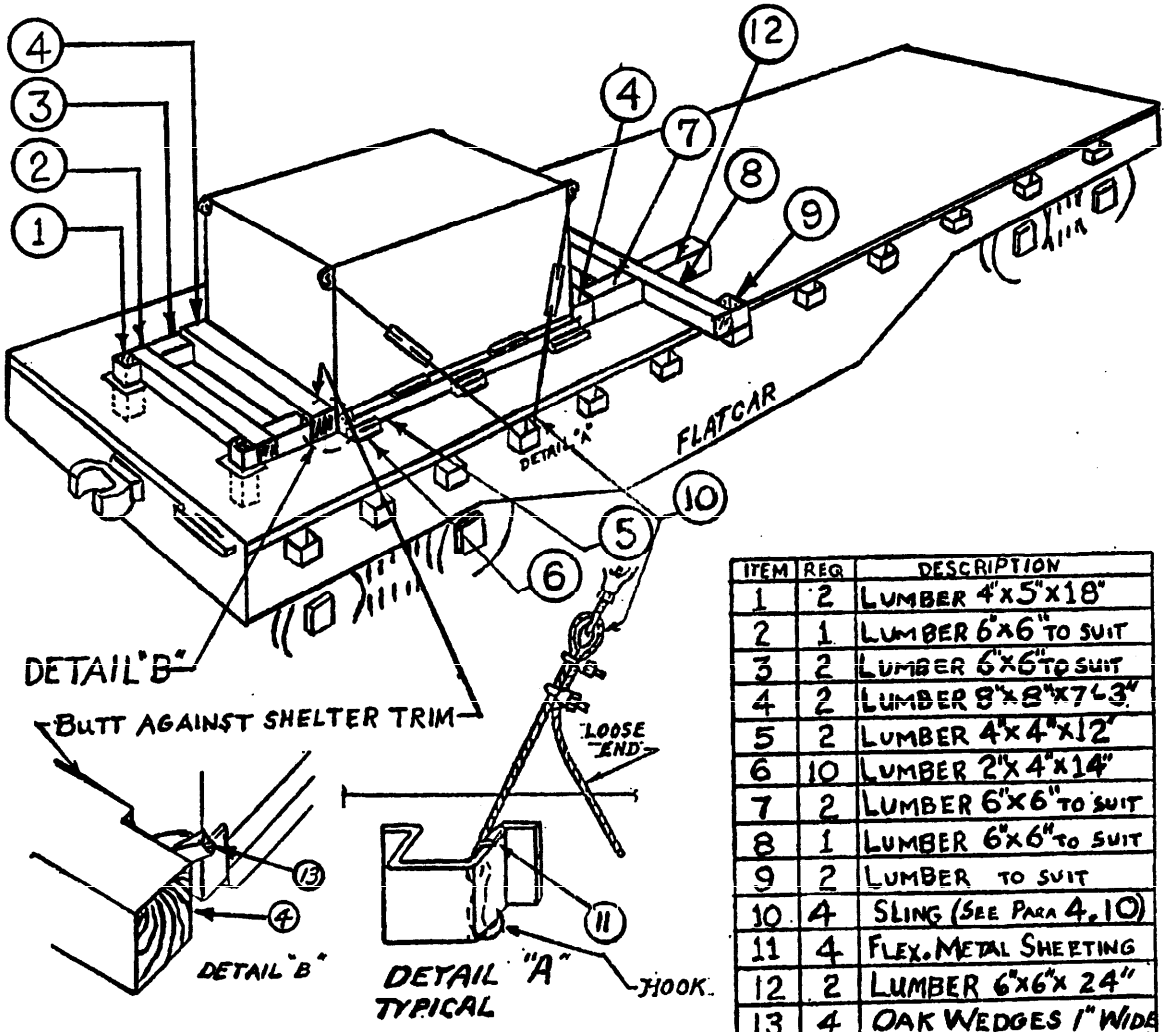
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4.9.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. Each timber shall be anchored down securely by any effective means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- The shelter facility loaded as specified in paragraph 4.9, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown slings (contractor furnished), shall be as specified in figure 5 and shall meet the provisions of the test as specified in paragraph 4.10.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds, traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative), shall be impacted against the stationary loaded test car coupled to the two empty flatcars, all with brakes off. All cars shall have standard (rigid) draft gear. The test shall consist of four impacts, two from each end of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the four sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8"x7'-3"
5	2	LUMBER 4"x4"x12'
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	4	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING
12	2	LUMBER 6"x6"x 24"
13	4	OAK WEDGES 1" WIDE

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 13 between towing corner forging and item 4 notch (Detail B)
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 5

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the figure referred above. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9, shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc., shall be subjected to the following two part test: (NOTE: Perform tests with inside cover of entrance boxes removed).

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a. Each of the five exposed surfaces of th shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches.

All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See figure 6 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

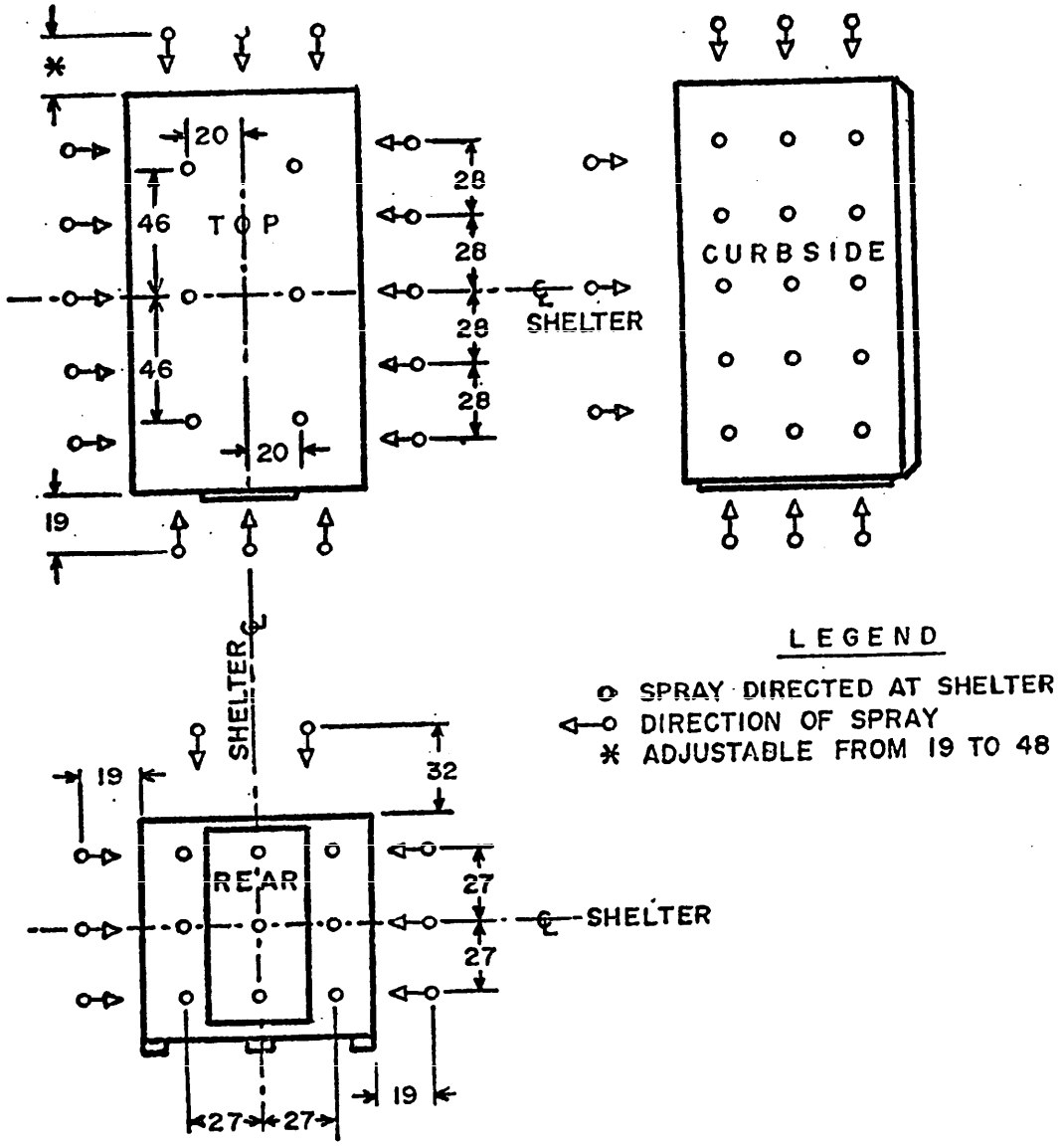
(NOTE: For first article and Group C tests only, the blowers and housings with exhaust vent closed shall be mounted outside the shelter in the operating position. Group B tests shall have blowers and housings in the storage position and the openings secured with the exhaust blower covers.)

b. With exhaust blowers and housings in their storage position, the shelter openings secured with the exhaust blower covers, and all other covers opened to their operating position (except the shelter door vent panel), repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.14 Mounting insert tests.-

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 6
PAGE 50

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4.14.1 First article and group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table VI. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V - Mounting Inserts Loads

Insert Size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in lbs)	Axial(lbs)	Torque(in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

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b. Size and mounting dimensions of racks and cabinets.

4.17 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

TABLE VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. 4. Sealer improperly applied (holes, separations, or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc., inoperative. 9. Electrical receptacles, switches, jacks, connectors, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16". 12. Center pin of rf cable assemblies improperly crimped or soldered, loose, or protruding beyond outer edge of connector shell. 13. The blower housing does not align properly with the mounting inserts of the shelter facility.
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges on bracket not removed. 5. Screws loose. 6. Gaskets, not completely and securely adhered.

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4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7 Power, Signal, and Video Wiring

Paragraph 4.17 Visual and Mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with MIL-P-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-338( )/TRC is the shelter facility of the AN/TRC-121( ). The shelter facility is used to house the following operational equipments forming the Radio Terminal Set:

a. 2 ea AN/GRC-143( ) Radio Set

b. 1 ea Drawer No. 1 thru Drawer No. 5 and Drawer No. 6B Spare Modules.

c. 2 ea CV-425( )/U Converter, Telephone Signal

d. 2 ea H-156( )/U Handset

e. 2 ea TA-312( )/PT Telephone

f. 1 ea LS-147( )/FI Intercom

g. 1 ea PP-4763( )/GRC Power Supply

h. 1 ea AN/GRC-106( ) Radio Set

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads or the government furnished live loads (radio components) as specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-338( )/TRC-121.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-338R/TRC-121. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 5820-A511

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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